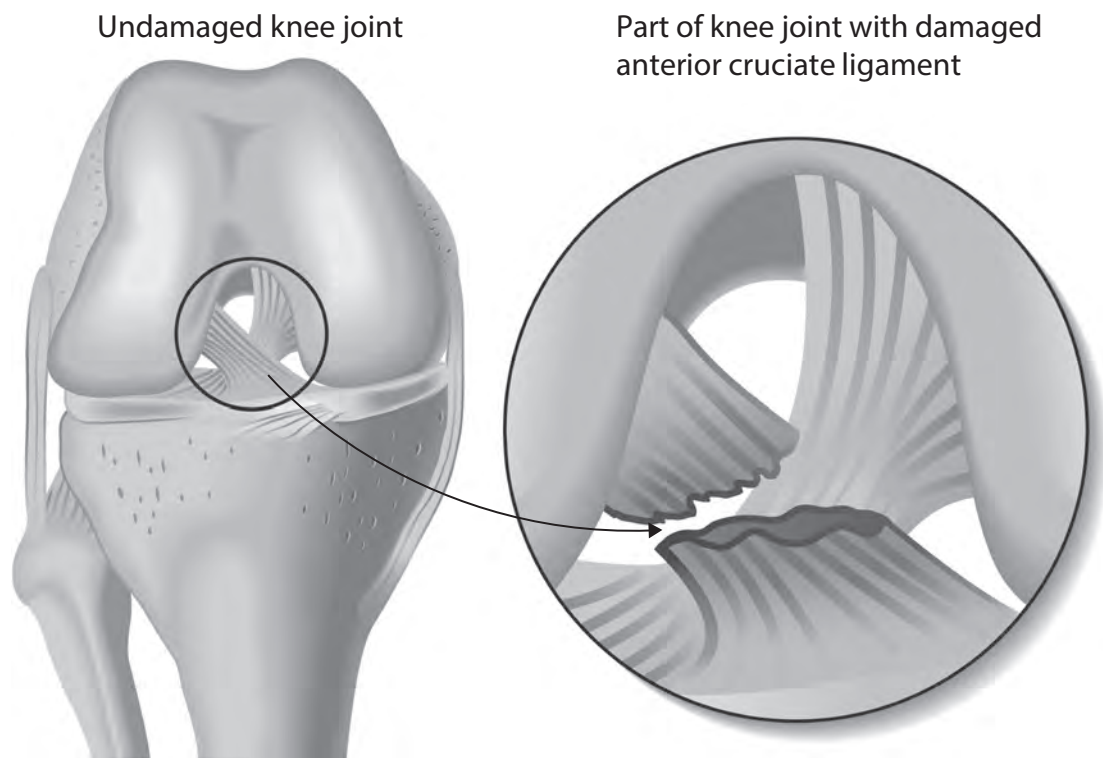


1 Muscles, tendons and ligaments interact to enable movement.

- (a) The diagram shows an undamaged knee joint and part of a knee joint with a damaged anterior cruciate ligament.



(Source: © rob3000/Alamy Stock Vector)

- (i) Which structure would the anterior cruciate ligament be attached to?

(1)

- ☐ **A** bone
- ☐ **B** cartilage
- ☐ **C** muscle
- ☐ **D** tendon

- (ii) Damage to anterior cruciate ligaments can be detected using magnetic resonance imaging (MRI).

Describe how MRI can show damage to the anterior cruciate ligament.

(2)

- (iii) In one country, there were 77 anterior cruciate ligament repairs for every 100 000 males in one year.

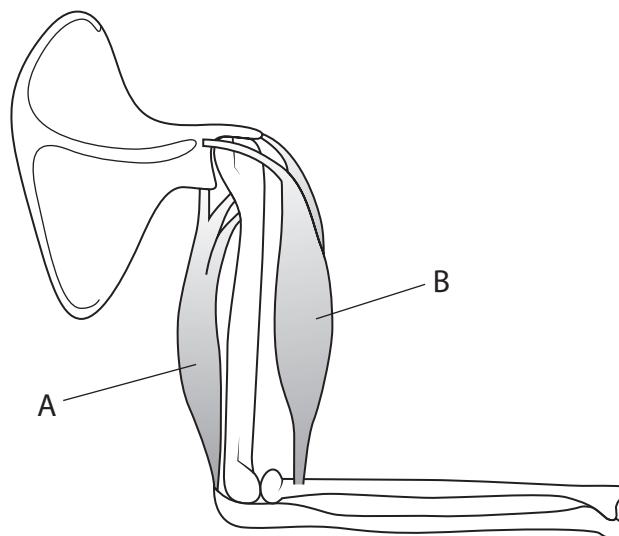
The population of this country in that year was 67 000 000 and 56% of the population were female.

Calculate the number of anterior cruciate ligament repairs in males that year.

(2)

Answer

(b) The diagram shows some muscles in an arm.



- (i) Name the **two** types of skeletal muscle that are contracted when the arm is held in this position.

(1)

- (ii) Describe how the arm could be straightened by the muscles labelled A and B.

(2)

(Total for Question 1 = 8 marks)